

Scored Non-Deterministic Finite Automata Processor for Sequence Alignment

Ryan Karbowniczak and Rasha Karakchi

Motivation

The rapid expansion of symbolic data in areas like internet data, biological data, and financial data has significantly increased the demand for efficient pattern matching.

Customized FPGA-based pattern matching engines such as NAPOLY are capable of efficiently reporting exact matches with high performance.

However, certain applications, like DNA sequence alignment, optimal path in a graph, or identifying the most likely pattern, require determining the best match.

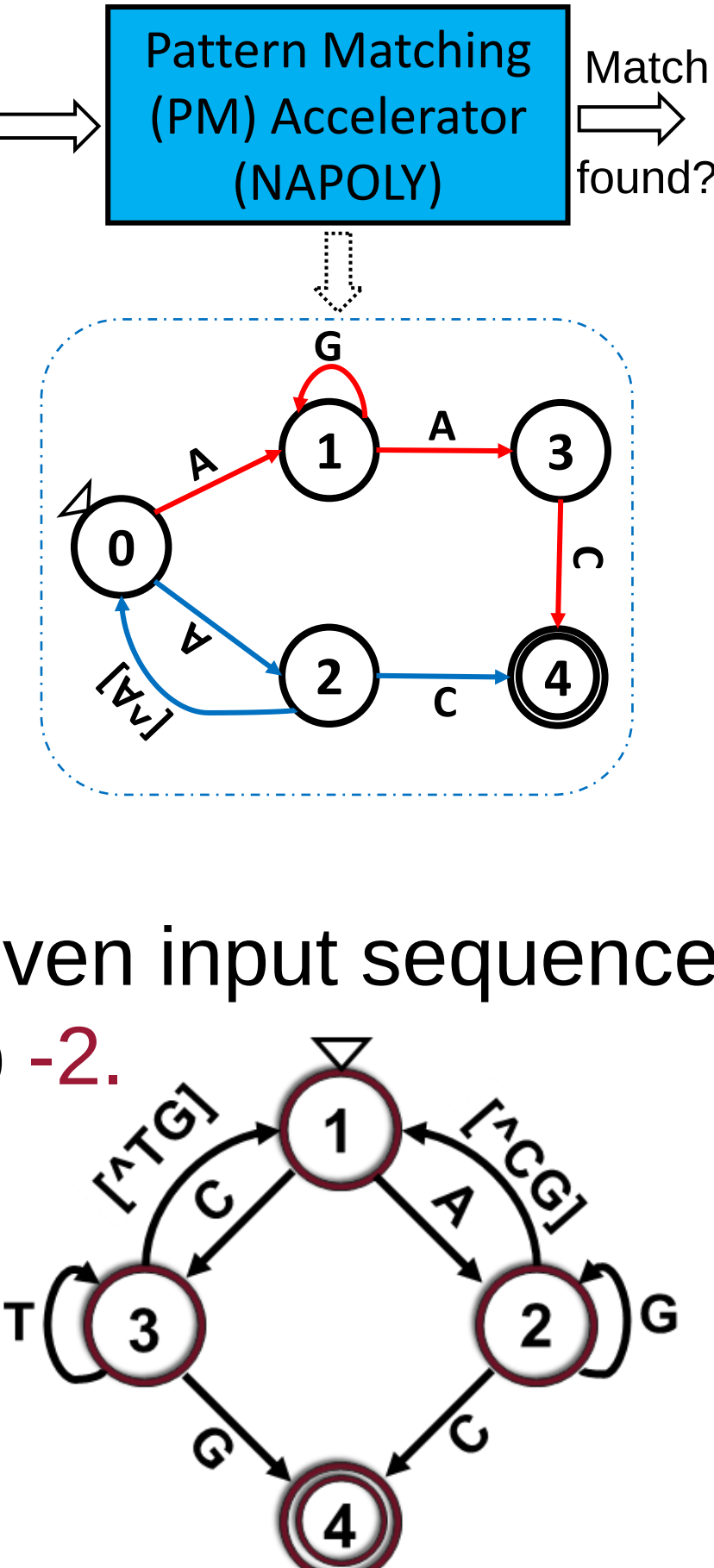
Example: Finding the highest score for the given input sequence graph, assuming match +2, mismatch -1, gap -2.

For the input sequence AGAC,

Highest score =

$$+2(1)+2(2)-1(1)+2(2) = 5$$

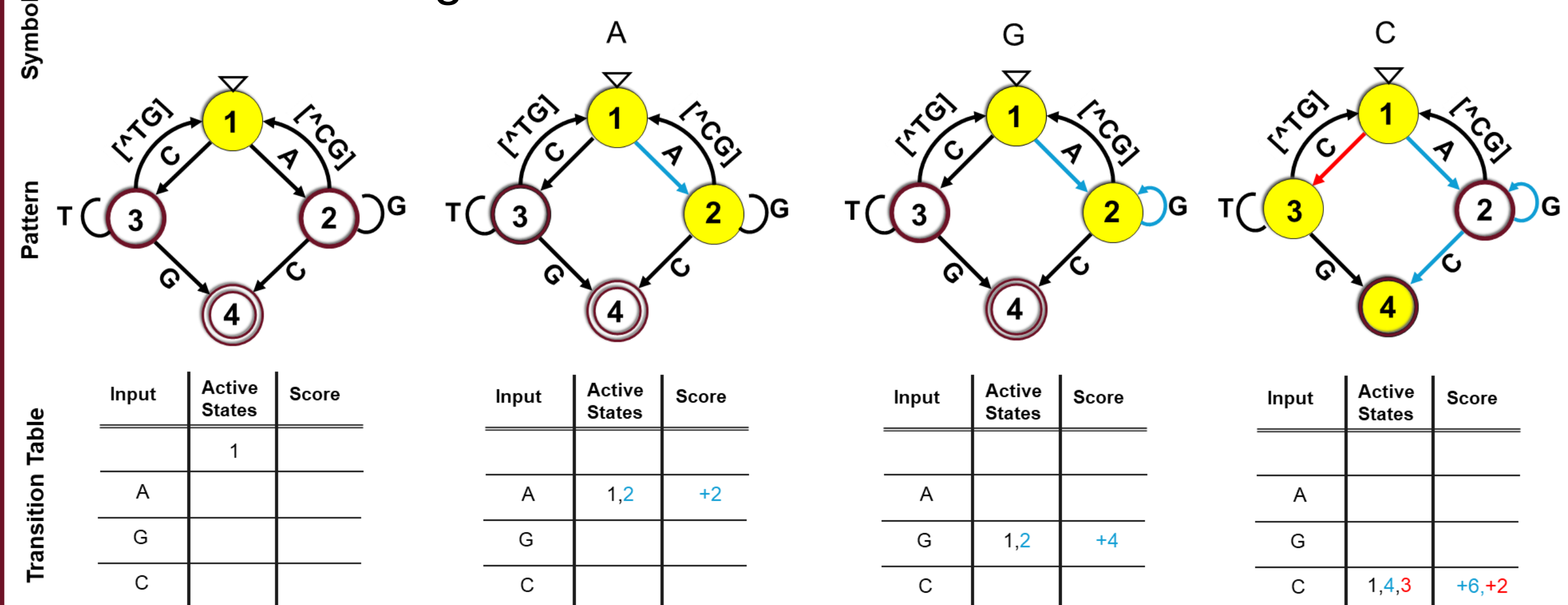
(1 mismatch, 3 match)



Challenges

NAPOLY, a FPGA-based non-deterministic automata processor

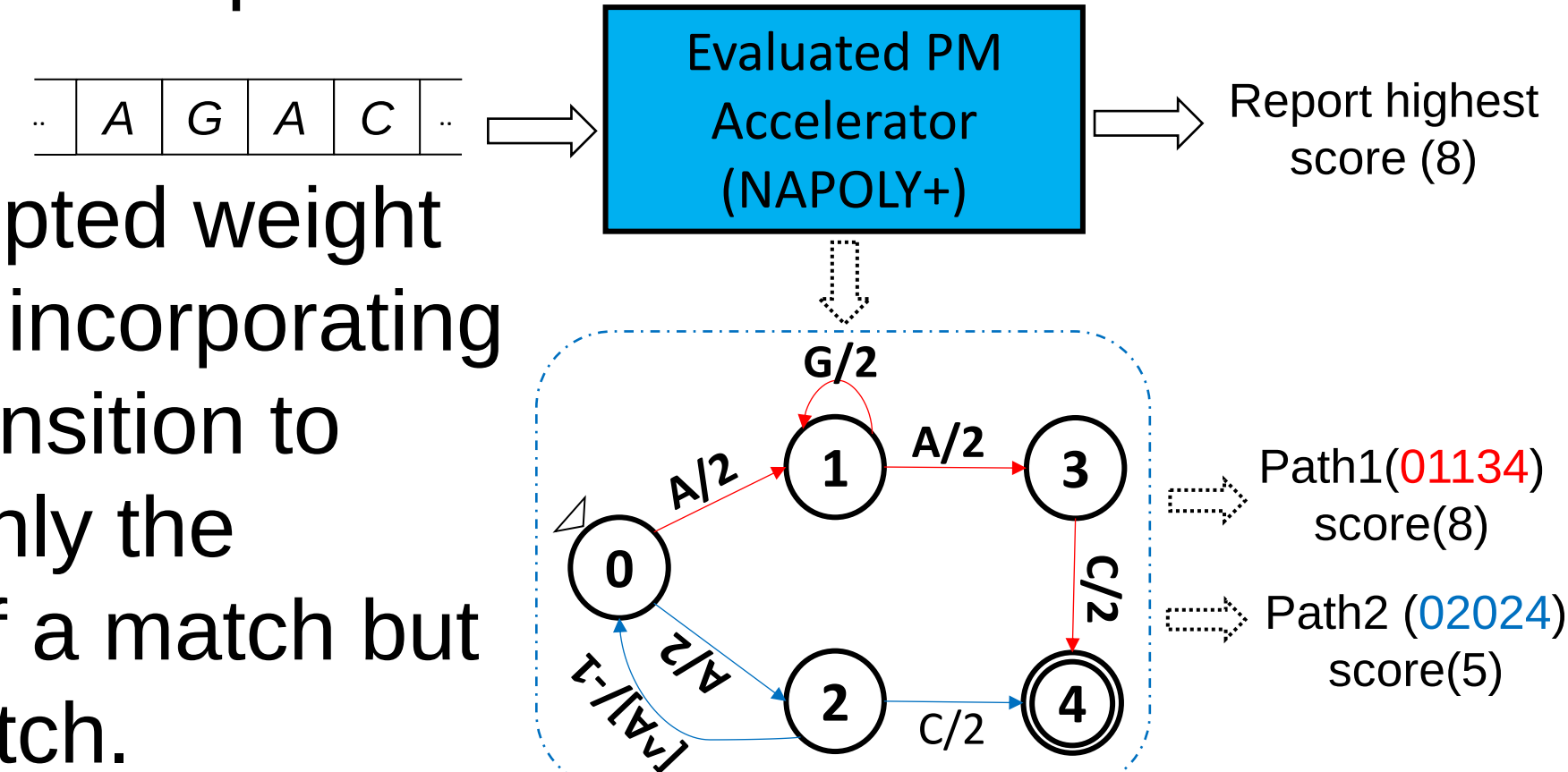
- Challenges:**
- Start state is active all time
 - Multiple states can be active simultaneously.
 - Track all possible paths and calculate the score for each path and find the highest score.



Contributions

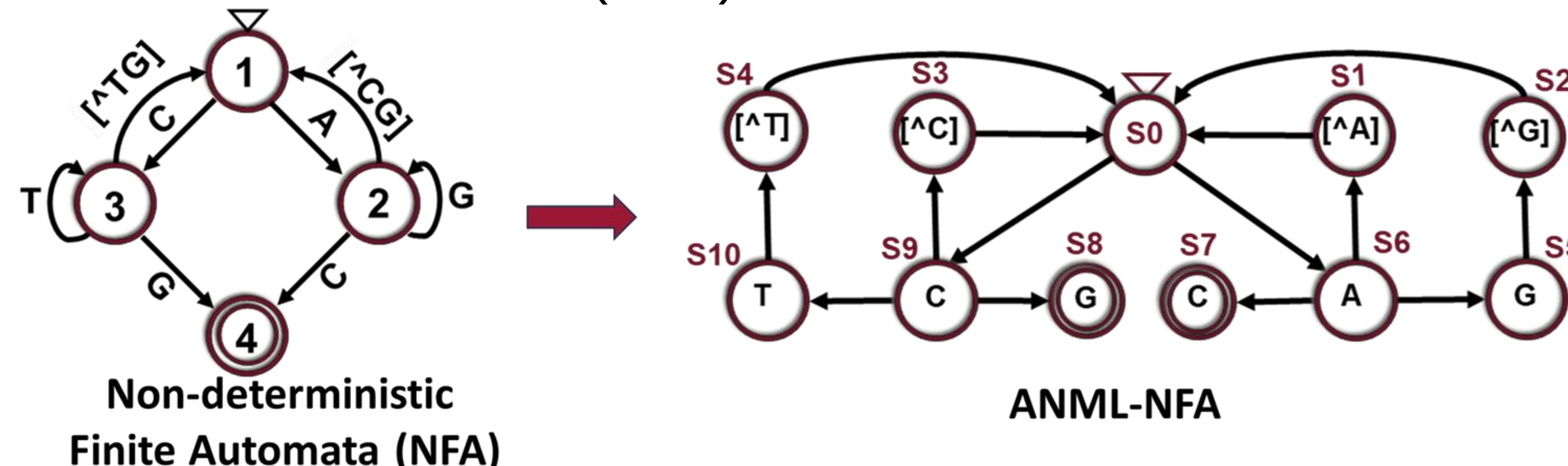
Evaluating the quality of the output of Pattern Matching NAPOLY accelerator to find the best sequence alignment by generating the highest score for DNA sequence match

To achieve this, we adopted weight automata processor by incorporating the scores for graph transition to allow considering not only the presence or absence of a match but also the quality of a match.

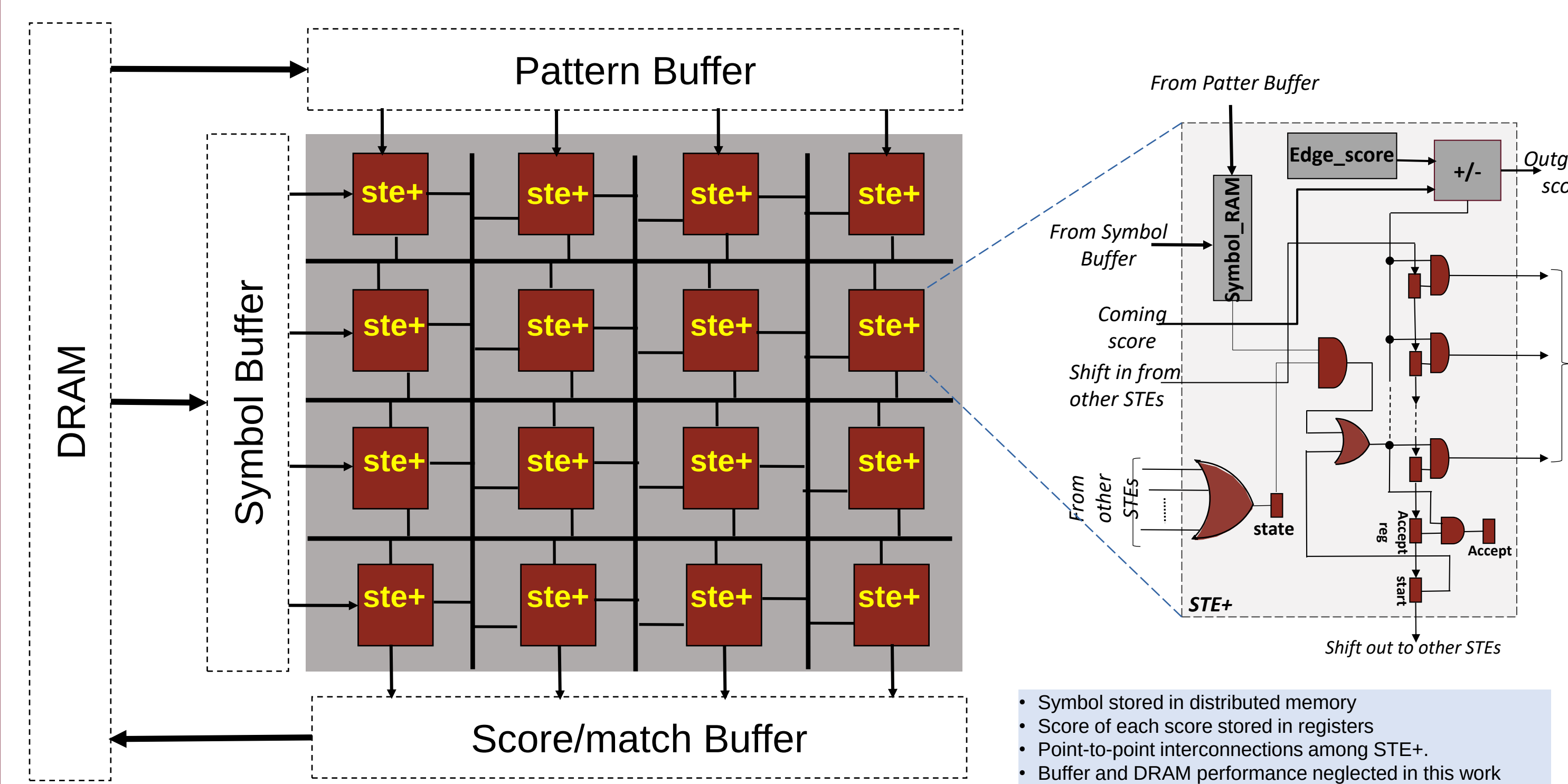


Proposed Design

NAPOLY designed based on alternative Micron-form of NFA description known as ANML (Automata Network Markup Language) associates transition labels with states instead of edges. Each state is called a State Transition Element (STE)

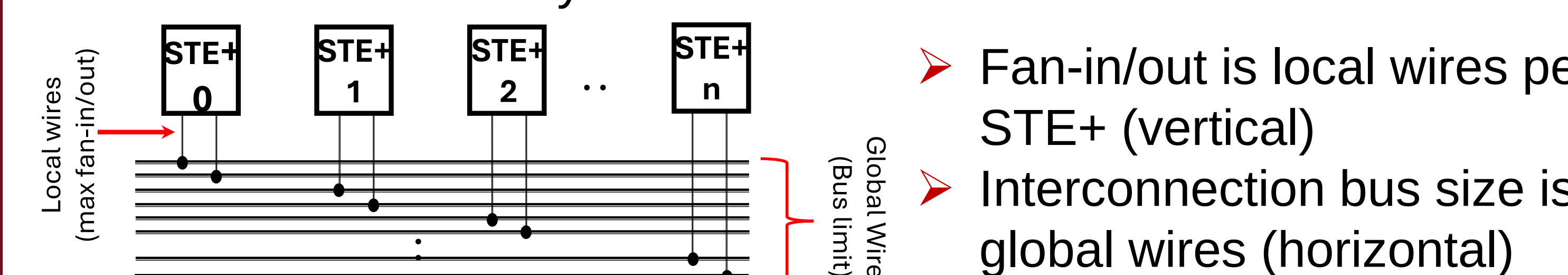
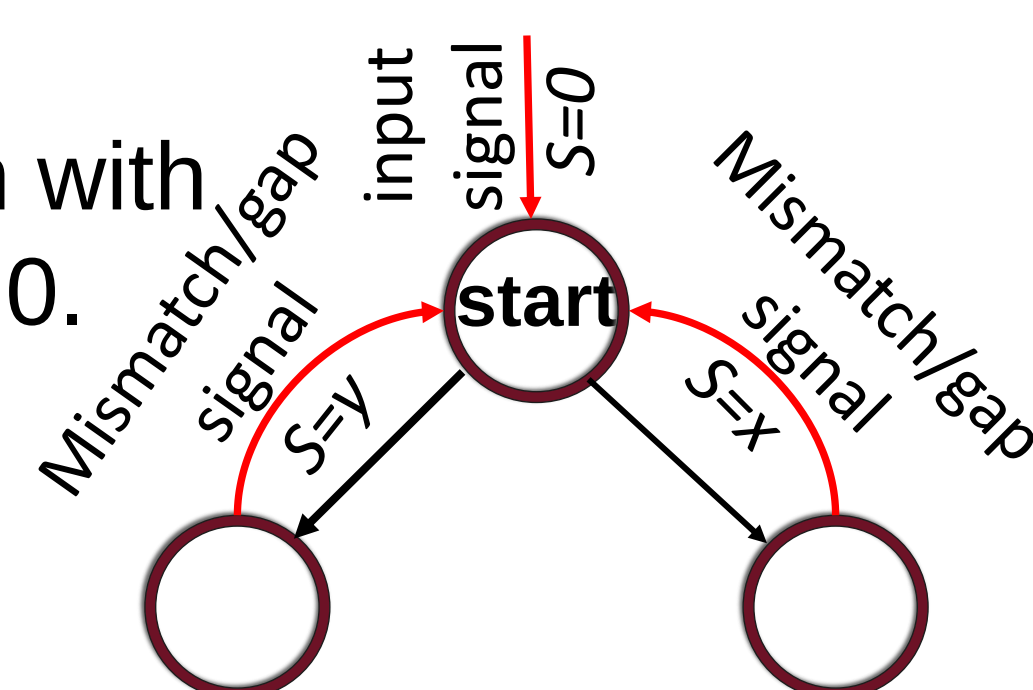


Proposed design consists of two-dimensional array of state transition elements (STE+) which incorporate symbol, score (weight), and arithmetic component to calculate the temporary score and find the final accumulated score



Design Limitations

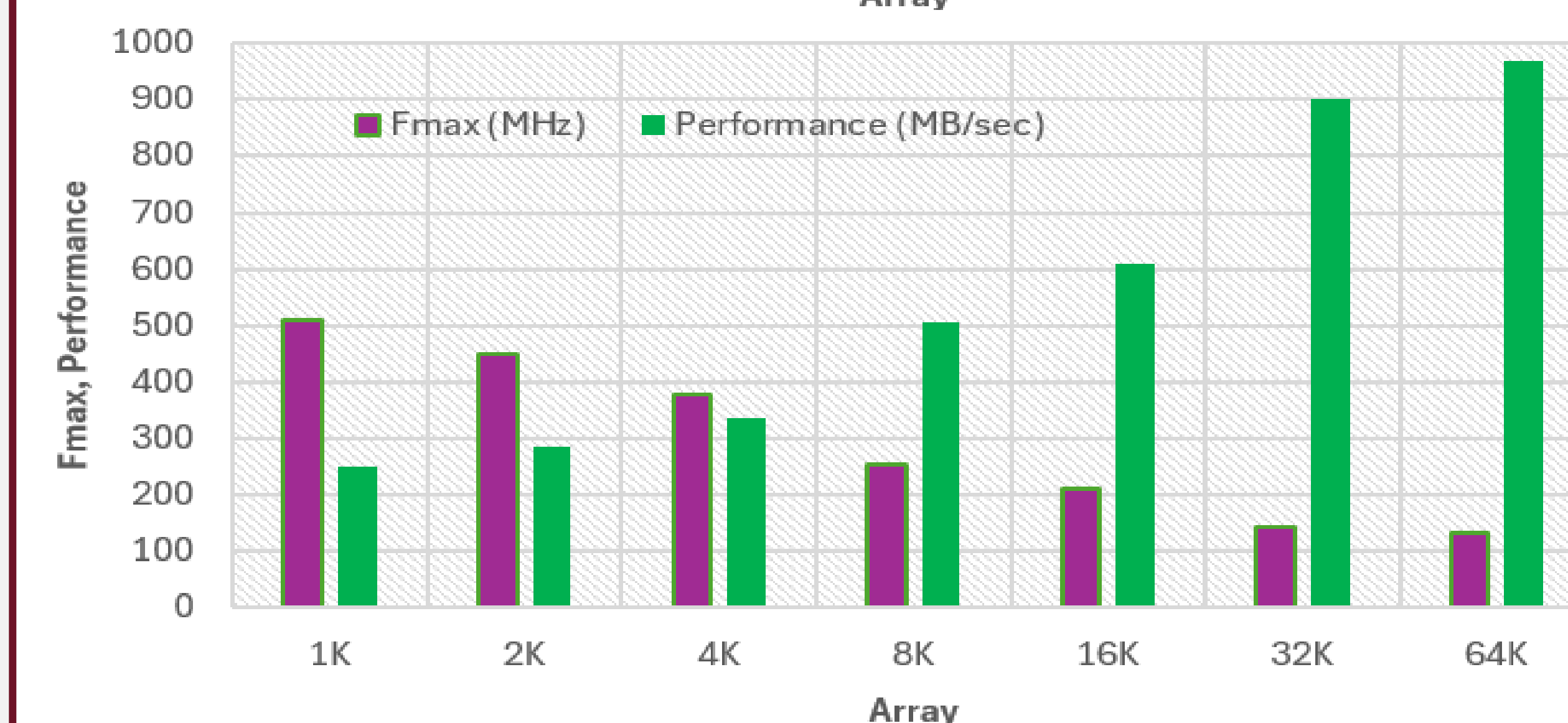
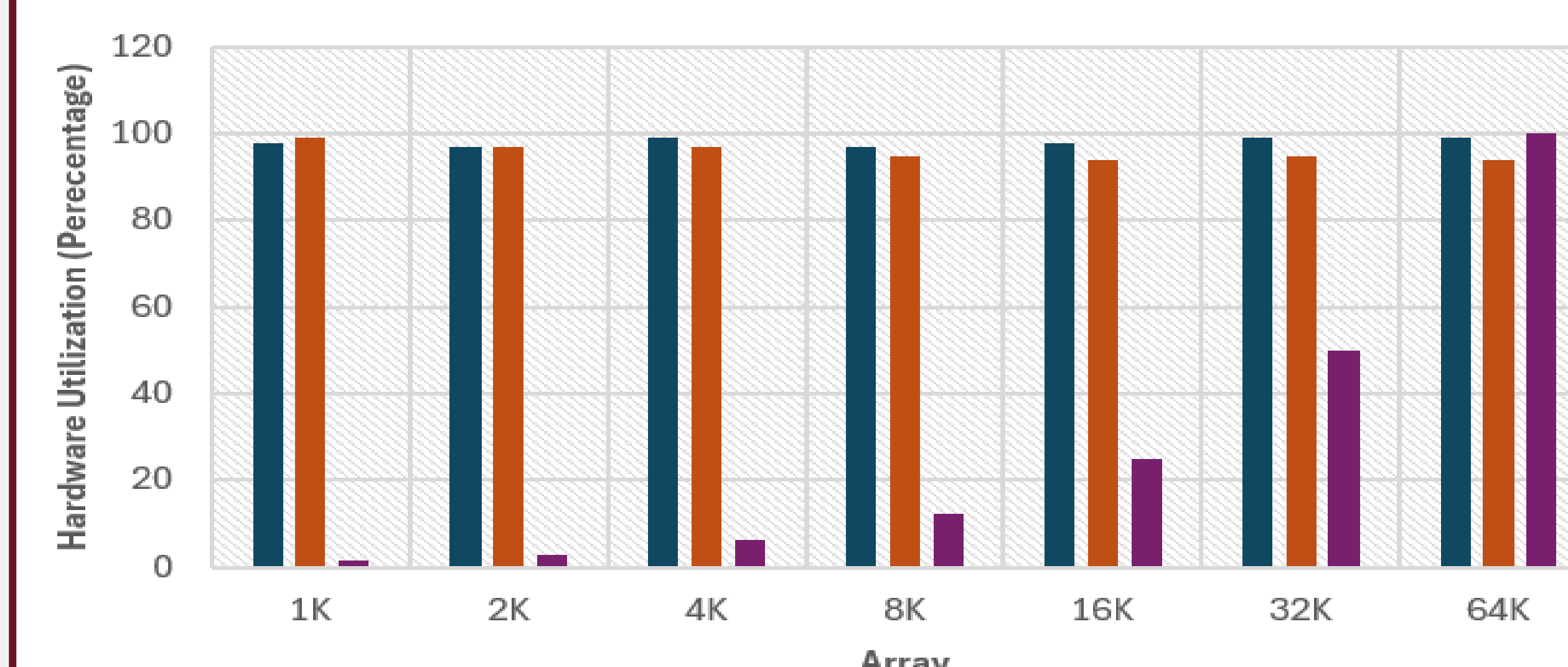
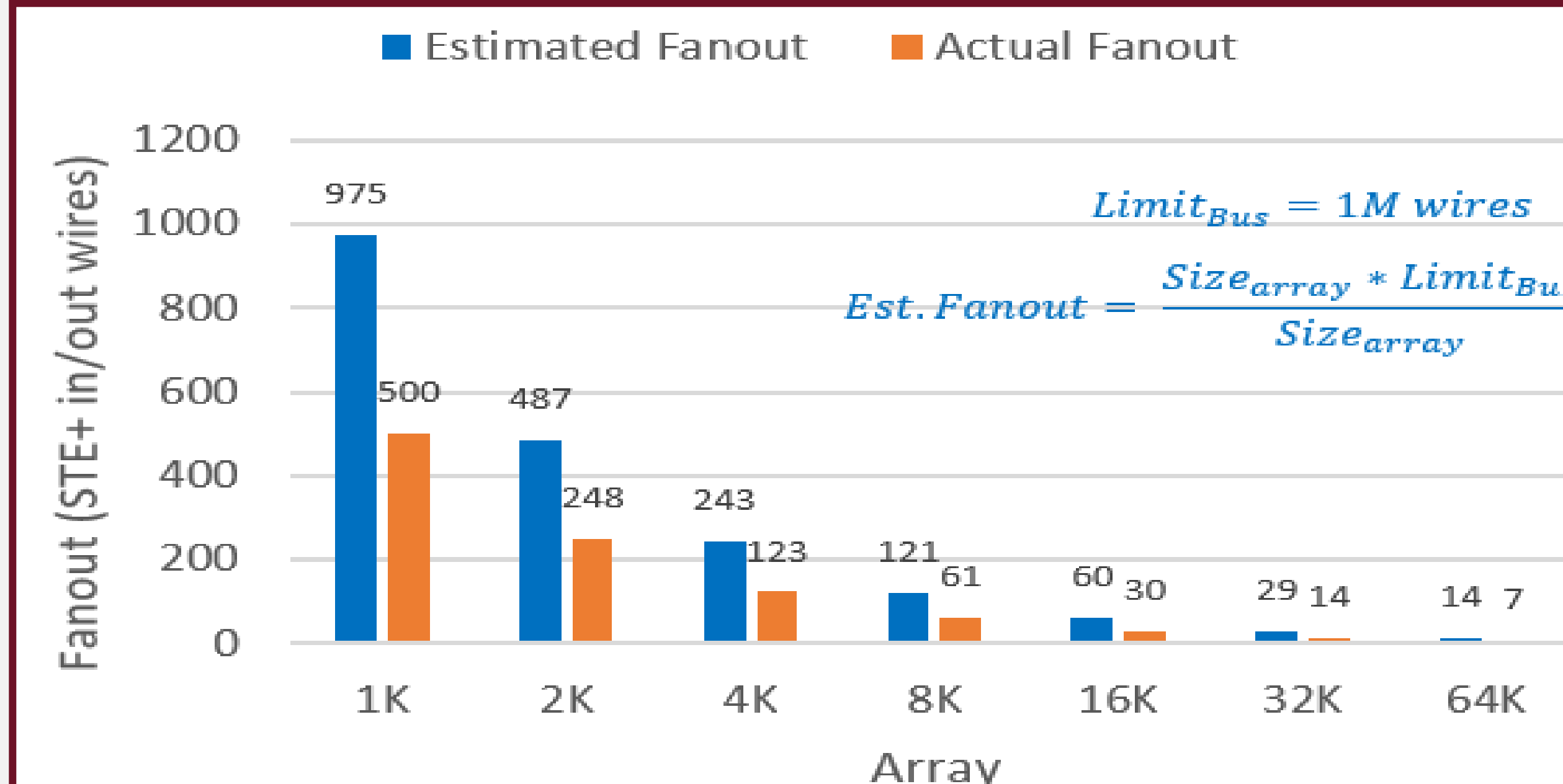
- To accommodate mismatch and gap, each STE+ connected to the Start state except reporting states
- Each new input symbol is a potential pattern with a score. Initialize any new symbol_score by 0.
- Number of incoming signals to start state limited by the max fan-in (incoming wires), limited by hardware resources
- Est. max fan-in/out limited by bus size (1000,000wires).
- Actual wires limited by overall available resources.



Evaluation Setup

- Tested the design on Zync 104 Ultrascale FPGA board.
- Total LUTs = 230400, total registers = 460800, Dist. Mem = 64KB

Results



Array size
versus maximum Fan-out

Hardware Utilization
For 7 configurations

Fmax, Performance
for 64K input symbol

Conclusion and Future Work

- We presented NAPOLY+, an automata processor overlay that has the capability to report the highest score among multiple sequence matches.
- Future work encompasses evaluating the design on actual dataset and develop a compilation software tool to map the logical datasets on the physical components

References and Acknowledgement

- Bradshaw, Jordan A., Rasha Karakchi, and Jason D. Bakos. "Two-Hit Filter Synthesis for Genomic Database Search." 2016 IEEE 24th Annual International Symposium on Field-Programmable Custom Computing Machines (FCCM). IEEE, 2016.
 - Yu, Xiaodong, et al. "Robotomata: A framework for approximate pattern matching of big data on an automata processor." 2017 IEEE International Conference on Big Data (Big Data). IEEE, 2017.
 - Karakchi, Rasha, and Jason D. Bakos. "NAPOLY: A Non-deterministic Automata Processor OverLaY." ACM Transactions on Reconfigurable Technology and Systems 16.3 (2023): 1-25.
 - Karakchi, Rasha, Charles Daniels, and Jason Bakos. "An overlay architecture for pattern matching." 2019 IEEE 30th International Conference on Application-specific Systems, Architectures and Processors (ASAP). Vol. 2160. IEEE, 2019.
- This work is supported by McNair Junior Fellowship at USC. Student travel was partially supported by the USC Office of Undergraduate Research Magellan Voyager travel award and by SC24 ACM SRC.